

Assignment 4: Vital Sign Description

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Vital sign chosen: Pulse

Pulse is the number of times the heart beats in one minute. It gives triage staff a quick view of how hard the cardiovascular system is working. I chose Pulse because it appears in the Week 0 dataset and I had already cleaned that column, so I could connect the write-up to the data work.

Clinical range and caveats

For most resting adults, a pulse from 60 to 100 beats per minute is treated as the usual range. A pulse below 60 bpm is called bradycardia, and a pulse above 100 bpm is called tachycardia. Those labels do not settle the clinical question by themselves. A fit adult may have a low resting pulse and be well. A patient with fever, pain, dehydration, bleeding, sepsis, anxiety, or shock may have a high pulse because the body is trying to compensate.

Triage priority

In triage, Pulse helps decide how quickly a patient needs reassessment, especially when it does not fit the rest of the picture. A fast pulse with low blood pressure, fever, confusion, chest pain, shortness of breath, or poor oxygenation should raise concern. A slow pulse matters more when the patient has dizziness, fainting, chest pain, weakness, or altered mental status. Pulse alone does not diagnose the cause, but it can flag a patient who needs earlier clinical review.

How I would use it

I would compare Pulse with blood pressure, respiratory rate, temperature, oxygen saturation, symptoms, medication history, and the patient's baseline fitness. I would also repeat the measurement if the value looks out of place, because movement, stress, poor technique, or data entry can distort the number. That keeps the value useful without treating one reading as the whole story.

References

MedlinePlus, Vital signs: <https://medlineplus.gov/ency/article/002341.htm>

American Heart Association, Target Heart Rates Chart:
<https://www.heart.org/en/healthy-living/fitness/fitness-basics/target-heart-rates>